Functions & Graphs

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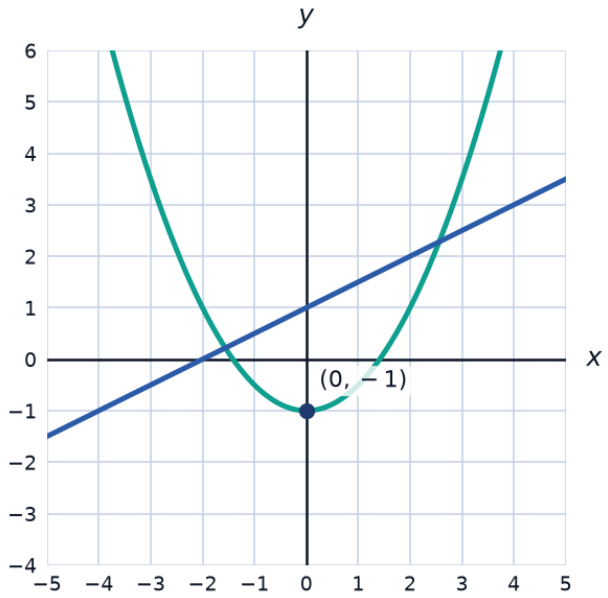
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating functions

If $f(x) = 3x - 2$, find $f(4)$.

2 [Foundational] • Target: Evaluating functions

If $f(x) = x^2 + 2x$, find $f(-3)$.



3 [Foundational] • Target: Evaluating functions

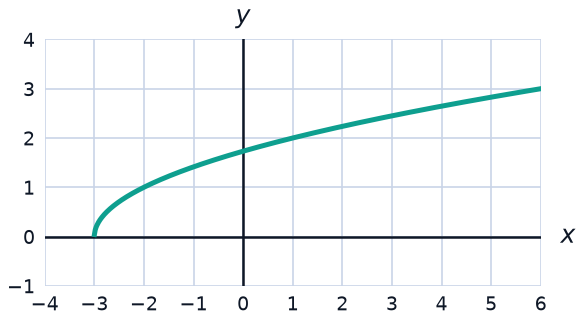
Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:

a $f(0)$

b $f(2)$

4 [Foundational] • Target: Domain and range

The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.



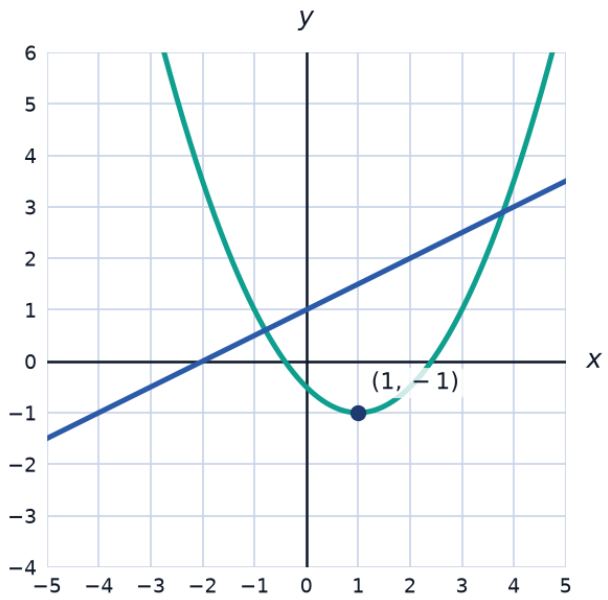
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5 [Foundational] • Target: Domain and range

Find the domain of $f(x) = \frac{1}{x-5}$.

6 [Foundational] • Target: Domain and range

Find the domain of $f(x) = \sqrt{2x-8}$.



7 [Foundational] • Target: Composition of functions

If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Composition of functions

If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.

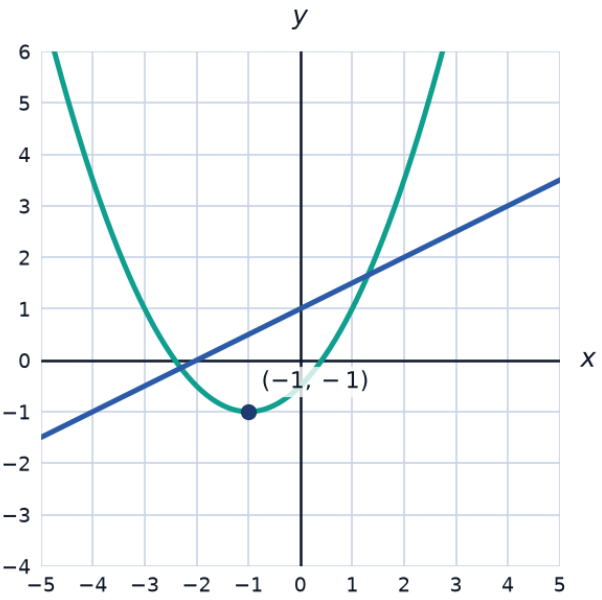
9 [Intermediate] • Target: Composition of functions

If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

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10 [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = 4x - 8$.



11 [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = \frac{x-3}{2}$.

12 [Intermediate] • Target: Inverse functions

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.

13 [Intermediate] • Target: Transformations of graphs

The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f.

14 [Intermediate] • Target: Transformations of graphs

If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

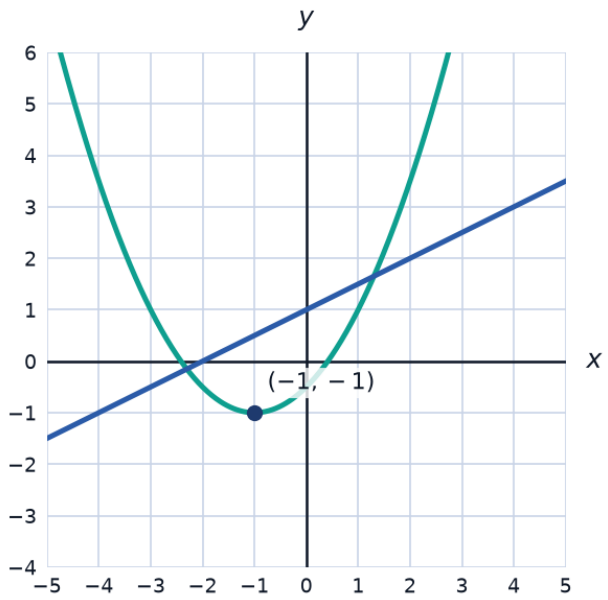
15 [Challenging] • Target: Piecewise functions • Cross-Lesson: 4.01 Ratios, Rates & Data

A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.

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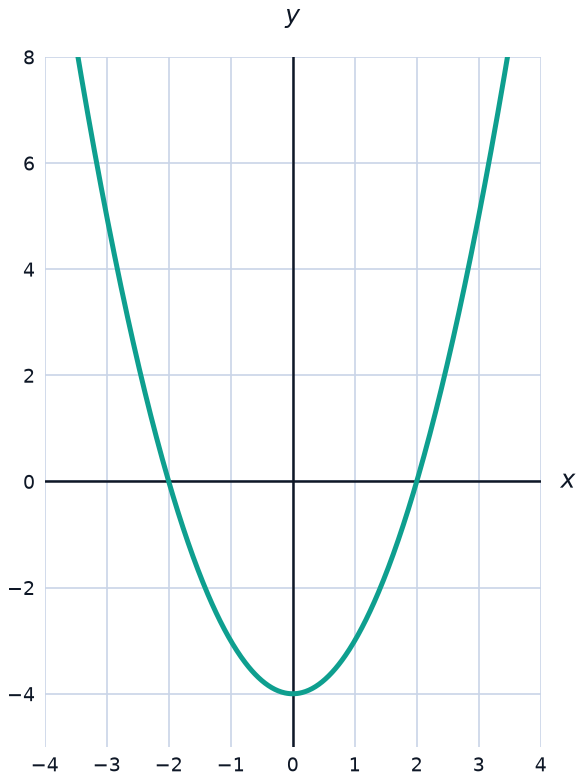
16 [Challenging] • Target: Piecewise functions • Cross-Lesson: 5.01 Statistics & Probability

A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.



17 [Challenging] • Target: Reading function graphs • Cross-Lesson: 6.01 Geometry

The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x-intercepts and y-intercept, then find them exactly.



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18 [Challenging] • Target: Even, odd, and other function properties • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.

19 [Challenging] • Target: Even, odd, and other function properties • Cross-Lesson: 8.01 Exponentials & Logarithms

Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

20 [Challenging] • Target: A well-designed word problem: modeling a car rental cost • Cross-Lesson: 1.01 Heart of Algebra

This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating functions

If $f(x) = -4x^2 + 3x - 4$, find $f(7)$.

a. -177

b. -179

c. -181

d. -11

22 [Foundational] • Target: Domain and range

If $f(x) = 3x^2 - 4x + 5$, find $f(6)$.

a. 87

b. 89

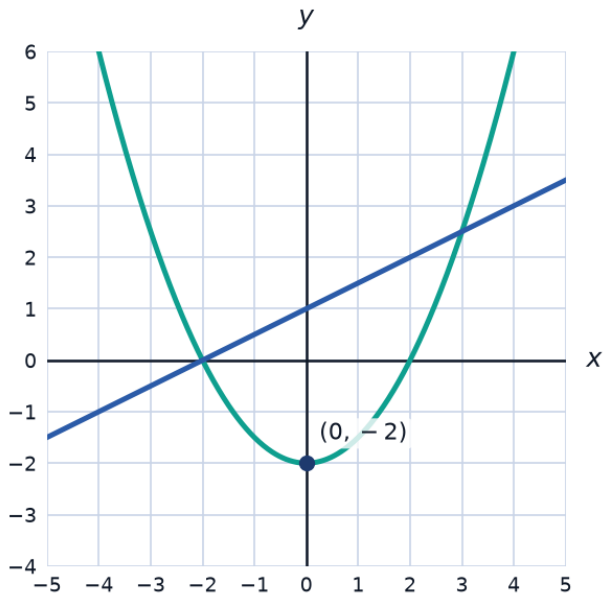
c. 91

d. -1

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23 [Foundational] • Target: Composition of functions

If $f(x) = x^2 - 9x$, find $f(8)$.



a. -64

b. -6

c. -8

d. -10

24 [Foundational] • Target: Inverse functions

If $f(x) = -4x^2 + 8x - 8$, find $f(4)$.

a. -42

b. -38

c. 8

d. -40

25 [Foundational] • Target: Transformations of graphs

If $f(x) = -4x^2 - 7x + 3$, find $f(8)$.

a. -309

b. -307

c. -311

d. -85

26 [Foundational] • Target: Piecewise functions

If $f(x) = -2x^2 - 3x + 1$, find $f(6)$.

a. -29

b. -87

c. -89

d. -91

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27 [Foundational] • Target: Reading function graphs

If $f(x) = -4x^2 + 3x + 7$, find $f(8)$.

a. -223

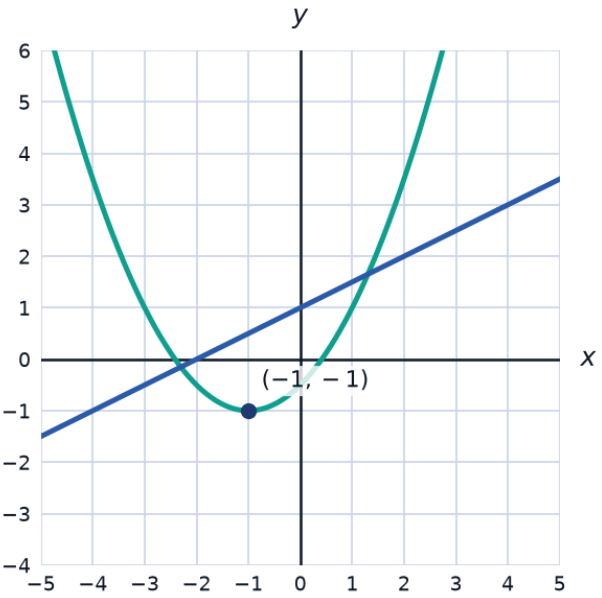
b. -227

c. -1

d. -225

28 [Foundational] • Target: Even, odd, and other function properties

If $f(x) = x - 2$ and $g(x) = 4x^2 + 3$, find $(f \circ g)(-6)$.



a. 148

b. 147

c. 145

d. 143

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: modeling a car rental cost

If $f(x) = 4x - 3$ and $g(x) = 2x^2 + 2$, find $(f \circ g)(0)$.

a. 5

b. 3

c. 7

d. 2

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30 [Intermediate] • Target: Function Notation & Operations

If $f(x) = -3x - 9$ and $g(x) = -4x^2 + 2$, find $(f \circ g)(-3)$.

a. 91

b. -34

c. 95

d. 93

31 [Intermediate] • Target: Composite Functions & Inverses

If $f(x) = -4x + 3$ and $g(x) = -2x^2 - 5$, find $(f \circ g)(-1)$.

a. 29

b. 31

c. 33

d. -7

32 [Intermediate] • Target: Graph Transformations

If $f(x) = -x + 3$ and $g(x) = 2x^2 + 3$, find $(f \circ g)(-2)$.

a. -8

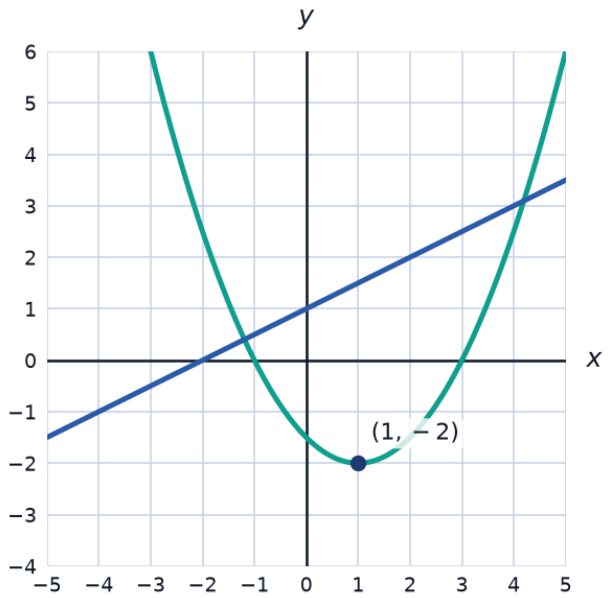
b. -10

c. 11

d. -6

33 [Intermediate] • Target: Rational Graphs & Asymptotes

If $f(x) = 3x - 4$ and $g(x) = -2x^2 - 2$, find $(f \circ g)(1)$.



a. -16

b. -18

c. -4

d. -14

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34 [Intermediate] • Target: Evaluating functions

If $f(x) = -x - 3$ and $g(x) = 3x^2$, find $(f \circ g)(0)$.

a. -1

b. -5

c. 0

d. -3

35 [Intermediate] • Target: Domain and range

Find the inverse of $f(x) = 3x + 4$.

a. $\frac{x-4}{-3}$

b. $\frac{x-4}{3}$

c. $\frac{x+4}{3}$

d. $3x + 4$

36 [Intermediate] • Target: Composition of functions

Find the inverse of $f(x) = 4x + 9$.

a. $\frac{x-9}{-4}$

b. $\frac{x+9}{4}$

c. $\frac{x-9}{4}$

d. $4x + 9$

37 [Intermediate] • Target: Inverse functions

Determine whether $f(x) = 2x^4 + 7x^2$ is even, odd, or neither.

a. both even and odd

b. neither

c. odd

d. even

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Transformations of graphs • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Determine whether $f(x) = -x^3 + 2$ is even, odd, or neither.

a. both even and odd

b. even

c. neither

d. odd

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39 [Challenging] • Target: Piecewise functions • Cross-Lesson: 8.01 Exponentials & Logarithms

A function is defined by $f(x) = -x$ for $x < -3$ and $f(x) = -5x - 5$ for $x \geq -3$. Find $f(-4)$ and $f(2).$

a. $6, -15$

b. $4, -17$

c. $4, -15$

d. $-15, 4$

40 [Challenging] • Target: Reading function graphs • Cross-Lesson: 1.01 Heart of Algebra

A function is defined by $f(x) = 2x + 1$ for $x < 0$ and $f(x) = -3x - 6$ for $x \geq 0$. Find $f(-5)$ and $f(3).$

a. $-9, -15$

b. $-9, -17$

c. $-7, -15$

d. $-15, -9$

41 [Challenging] • Target: Function Notation & Operations • Cross-Lesson: 2.01 Quadratics & Polynomials

If $f(x) = 4x - 5$ and $g(x) = 2x^2 + 1$, what is $(f + g)(3)$?

a. 24

b. 26

c. 28

d. 30

42 [Challenging] • Target: Function Notation & Operations • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the domain of the function $h(x) = \frac{\sqrt{x-4}}{x-7}$?

a. $[4, 7) \cup (7, \infty)$

b. $[4, \infty)$

c. $(4, 7) \cup (7, \infty)$

d. $x \neq 7$

43 [Challenging] • Target: Composite Functions & Inverses • Cross-Lesson: 5.01 Statistics & Probability

If $f(f^{-1}(x)) = x$ for all x, and $f(4) = 9$, what is $f^{-1}(9)$?

a. 4

b. $\frac{1}{4}$

c. $\frac{1}{9}$

d. 9

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44 [Challenging] • Target: Graph Transformations • Cross-Lesson: 6.01 Geometry

The graph of $y = x^2$ is vertically stretched by a factor of 3 and reflected across the x-axis. What is its equation?

a. $y = -3x^2$

b. $y = 3(-x)^2$

c. $y = -\frac{1}{3}x^2$

d. $y = -(3x)^2$

45 [Challenging] • Target: Rational Graphs & Asymptotes • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the vertical asymptote of $f(x) = \frac{3x+1}{x-5}$?

a. $x = 5$

b. $x = -5$

c. $y = 3$

d. $y = 5$

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